Journal of Machine Learning Research 1 (2026) 1–12 Submitted October 2026; Published

Towards Voice-Driven Financial Recordkeeping in Kingwana: Evaluating ASR Readiness for Small-Business Transactions in Bukavu, DRC

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Abstract

Small-business recordkeeping tools typically depend on typed interaction, while many everyday transactions are conducted orally in local language varieties. Our long-term objective is a Kingwana voice interface that can convert spoken business transactions into structured records and support voice-based database queries. This paper reports the first phase of that agenda: determining whether existing open-source ASR systems provide a reliable speech-recognition foundation for Kingwana transactional speech.

We evaluate Whisper Large v3, SunflowerASR-51, MMS-1B-All, OmniASR CTC 1B, and a Swahili-specific XLS-R model zero-shot on 500 scenario-guided natural utterances from 10 vendors in Bukavu, Democratic Republic of Congo. OmniASR obtains the lowest WER at 68.63%, followed by Sunflower at 70.51%, while Whisper reaches 117.70% with 2,813 insertions. In contrast, on a 2,124-clip Congolese Swahili TICO-19 medical read-speech set, WER ranges from 21.25% to 40.53%, revealing a large cross-corpus performance gap.

Application-oriented evaluation further changes the model ranking: Whisper preserves the most validated monetary mentions (55.49%), whereas OmniASR preserves the most currencies (26.00%), products/services (29.82%), and telecom operators (45.01%). A human review of 50 utterances across all five systems identifies frequent Kingwana-form errors and confirms an insertion-heavy Whisper failure mode. The results show that current systems are not ready for unattended financial recordkeeping and motivate a next phase that freezes the present business corpus as a held-out target-domain evaluation set, adapts models using additional Congolese Swahili and vendor speech, and ultimately evaluates Speech-to-Structured-Data and Speech-to-SQL.

Keywords: Kingwana, Congolese Swahili, automatic speech recognition, low-resource speech, African languages, domain-specific ASR, cross-domain evaluation, small-business speech

1 Introduction

The broader problem motivating this work is not speech recognition in isolation, but the interaction barrier between small-business owners and digital recordkeeping systems. In the setting that motivated this project, business information such as sales, customer debts,

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repayments, and inventory changes is naturally expressed through speech, while digital tools usually require the user to stop the transaction flow and type information into structured interfaces. Our long-term goal is therefore a Kingwana voice system through which a vendor can record or retrieve business information by speaking naturally, with the system converting speech into structured fields and database operations.

This goal creates a high-stakes speech-recognition problem. An ASR output can be partly readable yet still be unusable for financial recordkeeping if it changes an amount, currency, product, operator, customer name, or payment status. Prior work on speech-driven database querying demonstrates both cascaded and direct mappings from speech to SQL (Song et al., 2022, 2024; Liu et al., 2024), but such a pipeline depends on a sufficiently reliable speech representation. For an under-resourced regional variety such as Kingwana, that assumption must be tested rather than taken for granted.

Multilingual ASR has advanced through large-scale pretraining (Radford et al., 2023; Pratap et al., 2023; Babu et al., 2022), yet African languages remain unevenly represented and performance can vary across language varieties and domains (Caubrière and Gauthier, 2024; Imam et al., 2025; Nahabwe et al., 2026). We use *Kingwana* to refer to the Congo Swahili variety spoken by participating vendors in Bukavu. Instead of immediately fine-tuning a model, this Phase 1 pilot first establishes what existing systems can and cannot do on the target application speech.

Contributions. This paper makes four contributions. First, it introduces a 500-utterance, scenario-guided Kingwana small-business evaluation corpus from 10 vendors in Bukavu. Second, it benchmarks five open-source multilingual, African-adapted, and Swahili-specific ASR systems under a common zero-shot pipeline and contrasts the results with a 2,124-clip Congolese Swahili TICO-19 medical read-speech set (Anastasopoulos et al., 2020, 2021). Third, it moves beyond aggregate WER through business-field preservation, scenario/speaker analysis, reviewed code-switching, and 250 human-validated model–utterance judgments. Fourth, it uses these diagnostics to define the next development phases toward a Kingwana Speech-to-Structured-Data and Speech-to-SQL system rather than claiming that the present zero-shot models are deployment-ready.

2 Related Work

2.1 Multilingual and African-Language ASR

Whisper was trained on approximately 680,000 hours of multilingual and multitask weakly supervised speech (Radford et al., 2023); MMS expanded ASR coverage to more than 1,000 languages (Pratap et al., 2023); and XLS-R uses cross-lingual self-supervised pretraining across 128 languages (Babu et al., 2022). More recently, Omnilingual ASR expanded open multilingual recognition to more than 1,600 languages (Omnilingual ASR Team et al., 2025), while SunflowerASR-51 adapts Whisper Large v3 for 51 African languages, including Swahili (Hu et al., 2026).

Broad coverage does not imply equal reliability across regional varieties or applications. African ASR reviews highlight data scarcity, linguistic diversity, acoustic variability, and limited domain-specific resources (Caubrière and Gauthier, 2024; Imam et al., 2025). A benchmark across 13 African languages likewise shows that relative model performance

changes with architecture, data availability, and evaluation conditions (Nahabwe et al., 2026). These findings motivate direct evaluation on the speech distribution for which a system is intended.

2.2 Adaptation, Domain Shift, and Code-Switching

Targeted adaptation can substantially improve low-resource ASR. Bolarinwa et al. (2026), for example, report large WER reductions after fine-tuning Whisper Large v3 and MMS-1B-All on Yoruba–English code-switched speech. Our study deliberately precedes that step: we first freeze a zero-shot target-domain benchmark so that later adaptation can be measured against an untouched reference set.

Code-switching adds further difficulty because a single utterance can combine material from multiple languages (Mustafa et al., 2022). Kingwana business speech additionally includes regional forms, lexical borrowing, names, monetary expressions, and telecom vocabulary. We therefore separate literal transcription quality from application-critical field preservation and treat code-switch comparisons descriptively.

To test whether poor business-domain performance reflects a broader failure to recognize Congolese Swahili, we additionally evaluate the same systems on TICO-19 Congolese Swahili medical read speech. TICO-19 was created for COVID-19 communication in low-resource languages (Anastasopoulos et al., 2020); its Congolese Swahili speech data were also used in the IWSLT 2021 low-resource speech-translation evaluation (Anastasopoulos et al., 2021). Because this cross-corpus comparison changes not only domain but also speaking style, speakers, acoustics, and potentially model exposure to public data, it is interpreted as a distributional contrast rather than a causal estimate of domain effects.

2.3 From Speech Recognition to Structured Business Data

The intended endpoint of this research is not a transcript but an actionable business record. VoiceQuerySystem and Speech-to-SQL demonstrate cascaded speech-driven database querying (Song et al., 2022, 2024), while Wav2SQL explores direct speech-to-SQL parsing (Liu et al., 2024). For Kingwana, the prerequisite is to determine which speech information survives ASR reliably enough to support downstream structured extraction. This motivates the present emphasis on amounts, currencies, products, operators, payment methods, and qualitative error patterns in addition to WER and CER.

3 Materials and Methods

3.1 Bukavu Transactional Corpus

The target-domain corpus contains 500 manually transcribed recordings from 10 pseudonymous Kingwana-speaking telecom vendors working across 10 businesses in Bukavu, DRC. Participants included six men and four women, with 50 recordings per speaker. Five scenarios were balanced at 100 recordings each: sales, customer debt, inventory updates, debt repayment, and business-information queries.

Participants received semantic transaction prompts and responded naturally rather than reading fixed sentences. We therefore describe the material as *scenario-guided natural speech*. The standardized evaluation audio totals 3,631.420 seconds (approximately 60.5

Table 1: Summary of the target-domain Kingwana transactional corpus.

Property	Value
Speakers	10
Recordings	500
Recordings per speaker	50
Scenarios	5
Recordings per scenario	100
Standardized duration	3,631.420 s
Approximate duration	60.5 min
Normalized reference words	6,029
Human transcription coverage	500/500

minutes). All 500 clips were successfully decoded, manually transcribed, and aligned to structured annotations.

3.2 Cross-Corpus TICO-19 Evaluation

The same five systems were evaluated on a Congolese Swahili TICO-19 medical read-speech test set. The evaluation manifest contains 2,124 clips and 59,314 normalized reference words. TICO-19 originates from COVID-19/medical content (Anastasopoulos et al., 2020), and the Congolese Swahili speech was used as evaluation material in the IWSLT 2021 low-resource track (Anastasopoulos et al., 2021). We use this set as an external reference for cross-corpus generalization; it is not treated as a matched control for the Bukavu business corpus.

3.3 Transcription, Models, and Inference

Verbatim transcription preserved Kingwana lexical forms and observed French/English material rather than rewriting the references into Standard Swahili. The same deterministic normalization was applied to references and model predictions: Unicode NFC normalization, Unicode-aware lowercasing, punctuation replacement with spaces, whitespace collapse, and no semantic normalization of numbers. The business corpus used a single-annotator workflow; no independent inter-annotator agreement statistic is therefore reported.

Five open-source checkpoints were evaluated: Whisper Large v3, SunflowerASR-51, MMS-1B-All, OmniASR CTC 1B, and a Swahili-specific XLS-R model. All experiments were zero-shot with respect to the Bukavu corpus. Audio was standardized to mono 16 kHz PCM, and inference ran on NVIDIA L4 GPUs. Exact revisions, decoding settings, software versions, manifests, and reproduction commands are available in the project repository (Ashuza Albin et al., 2026).

3.4 Evaluation Measures

The primary word-level metric is

$$\mathrm{WER} = \frac{S + D + I}{N},$$

Table 2: ASR systems evaluated in both corpora.

Model	Role	Checkpoint
Whisper Large v3	General multilingual	`openai/whisper-large-v3`
SunflowerASR-51	African-adapted Whisper	`Sunbird/SunflowerASR-51-african-languages`
MMS-1B-All	Massively multilingual	`facebook/mms-1b-all`
OmniASR CTC 1B	Massively multilingual CTC	`facebook/omniASR-CTC-1B`
XLS-R Swahili	Swahili-specific	`thinkKenya/wav2vec2-large-xls-r-300m-sw`

where S, D, and I are substitutions, deletions, and insertions, and N is the number of reference words. CER is computed analogously over characters. We report corpus-level micro-WER/CER, speaker-macro WER, scenario-level WER, and word-level edit counts. Detailed per-speaker, noise/quality, and annotation breakdowns are available in the repository (Ashuza Albin et al., 2026).

For the business corpus, we additionally score strict preservation of researcher-validated, explicitly spoken amounts, currencies, products/services, telecom operators, and payment methods. This evaluates whether the ASR output itself preserves a field; it does not measure recovery by a downstream semantic model. Names and debt/payment terminology are assessed in the manual review rather than as corpus-wide entity accuracy.

For qualitative analysis, 50 business utterances were reviewed against all five model outputs, yielding 250 human-validated model–utterance judgments. The sample contains five clips per speaker and ten clips per scenario, with 34 deterministic background clips and 16 clips intentionally selected to inspect extreme Whisper insertion behaviour. Error labels are non-exclusive. A separate transcript-based review classified 85 business recordings as code-switched and 415 as not code-switched.

4 Results

4.1 Target-Domain and Cross-Corpus ASR Performance

Table 3 shows a large performance gap between the Bukavu transactional corpus and TICO-19 medical read speech. OmniASR is the strongest system on both sets, with 68.63% WER on business speech and 21.25% on TICO-19. Every model improves by at least 41.72 percentage points on the TICO-19 set. The contrast is descriptive: the two corpora differ in domain, speaking style, speakers, recording conditions, and potentially training-data exposure.

The cross-corpus comparison also changes the scale of Whisper's insertion behaviour. In the business corpus, Whisper produces 2,813 insertions over 6,029 reference words (46.7 insertions per 100 reference words). On TICO-19, it produces 7,351 insertions over 59,314 reference words (12.4 per 100). Thus, Whisper remains insertion-prone relative to the other systems, but the extreme target-domain failure pattern is substantially reduced on read medical speech.

Table 3: Cross-corpus ASR performance. ΔWER is business WER minus TICO-19 WER, in percentage points.

Model	Business WER	TICO WER	ΔWER	Business CER	TICO CER
OmniASR CTC 1B	**68.63**	**21.25**	47.38	**32.50**	**6.87**
SunflowerASR-51	70.51	28.79	41.72	35.96	15.66
MMS-1B-All	74.16	22.17	51.99	34.66	6.96
XLS-R Swahili	80.21	34.50	45.71	45.64	13.07
Whisper Large v3	117.70	40.53	77.17	72.34	17.18

Table 4: Word-level edit counts on the 500-clip business corpus.

Model	Sub.	Del.	Ins.	Ref. words
OmniASR CTC 1B	2,811	1,231	96	6,029
SunflowerASR-51	3,286	629	336	6,029
MMS-1B-All	3,478	695	298	6,029
XLS-R Swahili	3,549	1,186	101	6,029
Whisper Large v3	3,670	613	2,813	6,029

4.2 Target-Domain Error and Scenario Profiles

Within the business corpus, OmniASR makes the fewest substitutions and insertions but more deletions, while Whisper's total edit count exceeds the number of reference words. A WER above 100% is mathematically valid and should not by itself be labelled hallucination.

Performance also varies by transaction scenario. OmniASR obtains the lowest WER for customer debt, debt repayment, and sales, whereas Sunflower leads on business-information queries and inventory updates. The differences are not tested for statistical significance.

Speaker-macro WER is close to corpus WER for the main baselines (for OmniASR, 68.64% versus 68.63%), consistent with the balanced 50-clips-per- speaker design. Full per-speaker results are reported in the repository (Ashuza Albin et al., 2026).

4.3 Business-Critical Field Preservation

Application-oriented scoring produces a different ranking from aggregate WER. Whisper preserves the largest proportion of exact monetary mentions (55.49%) despite having the worst overall WER. OmniASR preserves the most currencies, products/services, and telecom operators, while Sunflower preserves the most payment methods. This shows why transcription metrics alone are insufficient for assessing financial-recordkeeping readiness.

4.4 Human Error Review and Code-Switching

Across 50 manually reviewed utterances per model, locally characteristic Kingwana-form errors appear in 46–49 outputs for every system. Acceptable orthographic/phonetic variants are also frequent, showing that literal string disagreement does not always correspond to a meaningful recognition failure. The review additionally identifies errors involving amounts, currencies, names, products/operators, debt/payment terminology, and French switching.

Table 5: Business-corpus scenario-level WER (%). Lowest value in each row is bold.

Scenario	Omni	Sunflower	MMS	XLS-R	Whisper
Business-information query	68.24	**67.22**	73.43	77.50	94.07
Customer-debt recording	**66.72**	72.09	73.53	79.66	150.04
Debt-repayment recording	**64.07**	67.05	65.90	75.74	90.47
Sales recording	**73.40**	76.36	81.71	83.95	121.81
Inventory update	70.85	**69.61**	76.55	84.15	132.62

Table 6: Strict preservation of researcher-validated, explicitly spoken business fields.

Model	Amounts	Currencies	Products/services	Operators	Payment methods
OmniASR	5/328 (1.52%)	**39/150 (26.00%)**	**102/342 (29.82%)**	**158/351 (45.01%)**	2/37 (5.41%)
Sunflower	124/328 (37.80%)	8/150 (5.33%)	50/342 (14.62%)	148/351 (42.17%)	**5/37 (13.51%)**
MMS	74/328 (22.56%)	37/150 (24.67%)	84/342 (24.56%)	129/351 (36.75%)	3/37 (8.11%)
XLS-R	0/328 (0.00%)	0/150 (0.00%)	6/342 (1.75%)	14/351 (3.99%)	0/37 (0.00%)
Whisper	**182/328 (55.49%)**	22/150 (14.67%)	56/342 (16.37%)	74/351 (21.08%)	4/37 (10.81%)

All 16 hallucination candidates occur in the deliberately selected high-insertion Whisper stratum; none occurs in the 34 background Whisper clips or in the other systems. Consequently, 16/50 is not a corpus-wide hallucination prevalence estimate. It demonstrates the existence of an insertion-heavy failure mode that warrants further targeted evaluation.

After transcript review, 85 business recordings are classified as code-switched and 415 as not code-switched. WER for code-switched versus not code-switched speech is 67.51/68.87 for OmniASR, 65.16/71.65 for Sunflower, 71.09/74.81 for MMS, 81.45/79.95 for XLS-R, and 207.53/98.49 for Whisper. The groups are unequal and potentially confounded, so these values do not establish a causal effect of code-switching.

Table 7: Counts of non-exclusive manual labels across 50 reviewed outputs per model. AMT: amount; CUR: currency; NAME: person name; PROD: product/operator; DEBT: debt/payment terminology; KING: Kingwana-form; FR: French switch; VAR: acceptable variant; SEM: semantic equivalent; HALL: hallucination candidate.

Model	AMT	CUR	NAME	PROD	DEBT	KING	FR	VAR	SEM	HALL
OmniASR	37	11	14	15	10	49	6	39	9	0
Sunflower	20	11	13	24	11	46	7	41	20	0
MMS	23	4	15	18	10	48	4	42	22	0
XLS-R	26	15	21	35	17	49	9	34	18	0
Whisper	21	14	22	30	16	48	11	32	7	16

5 Discussion

The central result is a target-application readiness gap. The zero-shot models are not uniformly incapable of Congolese Swahili: on TICO-19 medical read speech, WER falls to 21.25–40.53%, compared with 68.63–117.70% on the Bukavu business corpus. The same model, OmniASR, is best on both sets, but its WER increases by 47.38 percentage points in the target business setting. Whisper's insertion rate similarly falls from 46.7 to 12.4 inserted words per 100 reference words on TICO-19. The contrast is therefore consistent with a substantial distribution mismatch between public read-speech conditions and the target transactional setting. It does not isolate domain as the sole cause: speaking style, speakers, acoustics, transcription conventions, lexical content, and possible pretraining overlap may also contribute.

For financial recordkeeping, WER is necessary but not sufficient. The field-preservation results show that model rankings depend on what the application needs to preserve. Whisper has the highest overall WER yet the highest exact amount preservation, while OmniASR leads several other business fields. Conversely, OmniASR's omission-heavy profile may be particularly risky if a deleted token is an amount, currency, or payment status. This supports application-specific evaluation in which transcription quality is complemented by field accuracy and, eventually, structured-record and execution accuracy. The result is aligned with speech-to-database work where upstream recognition errors can propagate into downstream queries (Song et al., 2024; Liu et al., 2024).

The benchmark changes what the next engineering step should be. The present one-hour business corpus should now be treated as a frozen held-out target-domain evaluation set rather than consumed for opportunistic fine-tuning. The next phase should train or adapt promising systems on additional Congolese Swahili speech—for example, separate TICO-19 development material, Common Voice, other public Congolese Swahili speech, and newly collected speaker-disjoint vendor data—while preserving both the current business set and the TICO-19 test set for evaluation. OmniASR is a strong candidate because it leads both corpora; Sunflower is also valuable as an African-adapted generative baseline and preserves substantially more amounts than OmniASR. The goal of adaptation should not be only lower WER, but improved preservation of business-critical fields.

6 Limitations

The target-domain corpus is a pilot: 500 recordings, approximately 60.5 minutes of speech, 10 speakers, and one occupational/geographic setting in Bukavu. It is scenario-guided rather than unrestricted spontaneous conversation, so the findings should not be generalized to all Kingwana speakers or all Congo Swahili varieties.

Annotation used a single-annotator workflow, including the reviewed code-switching labels. The 50-utterance qualitative sample intentionally oversampled 16 extreme Whisper insertion cases, so its hallucination counts characterize a failure mode rather than population prevalence. Exact business-field preservation is also stricter than what a downstream semantic model might recover, and names/debt terminology were not scored corpus-wide.

Finally, the TICO-19 comparison is cross-corpus rather than controlled: medical read speech and transactional natural speech differ along several dimensions, and possible model exposure to public TICO-19 material cannot be ruled out. The study therefore establishes zero-shot readiness and a large distributional gap, not the best achievable Kingwana performance after adaptation or a causal estimate of domain effects.

7 Conclusion and Future Work

This paper establishes Phase 1 of a longer programme toward voice-driven financial record-keeping in Kingwana. Five open-source ASR systems were benchmarked on 500 Bukavu business utterances, with the best system, OmniASR, reaching 68.63% WER. The same systems perform substantially better on 2,124 TICO-19 Congolese Swahili medical read-speech clips, where OmniASR reaches 21.25% WER and MMS 22.17%. Together with the business-field and manual analyses, this indicates that the immediate bottleneck is not merely nominal support for Congolese Swahili, but robust transfer to the target transactional speech distribution and preservation of the fields needed for business records.

Phase 2: data expansion and adaptation. The current 500-clip business corpus will be frozen as the target-domain evaluation set. We plan to expand training data using external Congolese Swahili resources such as Common Voice (Ardila et al., 2020) and non-test TICO-19 material, while collecting additional vendor speech from new speakers. Fine-tuning and parameter-efficient adaptation will then be compared on speaker-disjoint data, with model selection based on both WER/CER and business-field preservation. The TICO-19 test set evaluated in this paper will remain held out rather than being reused for training.

Phase 3: speech to structured records. Once the ASR layer is sufficiently improved, the next experiment will map speech/transcripts to validated transaction fields such as amount, currency, product, operator, payment method, debt status, and customer information. Evaluation will move from transcription metrics to field-level precision, recall, exact-match structured records, and robustness to ASR errors.

Phase 4: Kingwana Speech-to-SQL and user evaluation. The long-term system will allow a vendor to say a transaction or ask a business question in Kingwana, automatically update or query the underlying database, and receive a usable response. Building on prior speech-driven SQL research (Song et al., 2022, 2024; Liu et al., 2024), we plan to evaluate SQL generation/execution accuracy and, ultimately, whether the voice workflow reduces

the practical burden of recording and retrieving business information for users in the target setting. The reproducible benchmark, configurations, and analysis artifacts are maintained in the project repository (Ashuza Albin et al., 2026).

Acknowledgments and Disclosure of Funding

We thank the participating small-business vendors for contributing their time and speech data to this pilot study.

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